



AI-BASED FOREST FIRE DETECTION & EARLY WARNING SYSTEM

A Technical Overview of Intelligent, Sensor-Driven Wildfire Surveillance

TECHNICAL PRESENTATION

Introduction

Why intelligent, early detection matters

Forest fires spread rapidly and unpredictably, often before ground personnel or manual reporting can respond. Traditional monitoring — watchtowers, periodic patrols, and citizen reports — reacts only after visible smoke or flame has already spread, by which time containment becomes far more difficult and costly.

An AI-based system continuously observes forest regions through satellite, aerial and ground sensor data, learns the visual and environmental signatures of fire and pre-fire conditions, and raises an alert within minutes — shifting the response model from reactive to predictive.



Delayed Detection

Manual monitoring identifies fires only after visible spread.



Vast, Remote Terrain

Forests span areas too large for constant human surveillance.

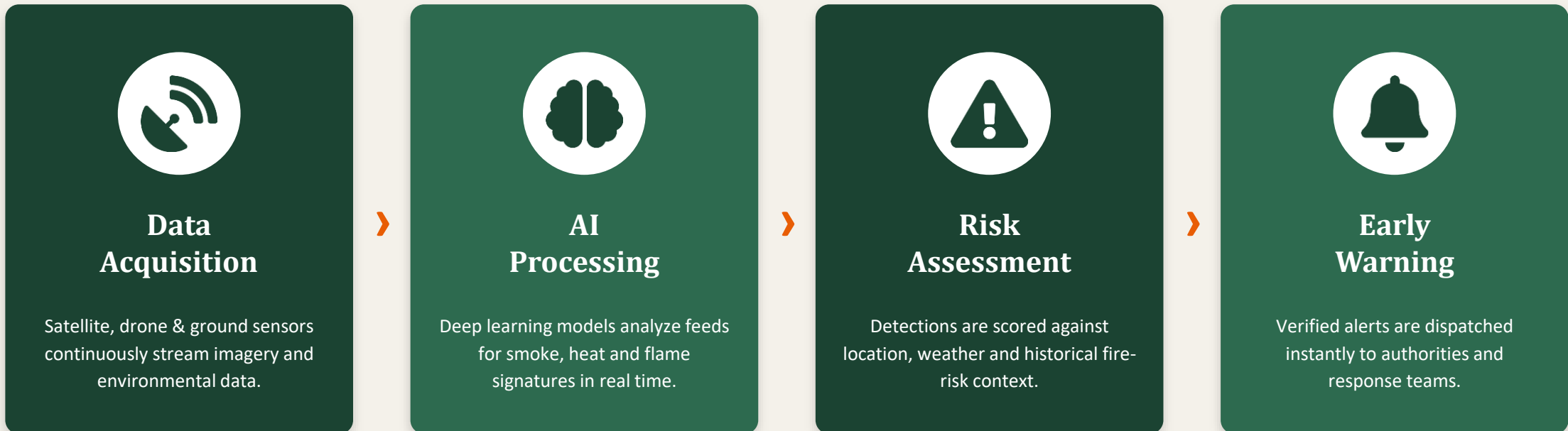


Rapid Fire Growth

Minutes of delay can multiply the area affected many times over.

System Architecture

A four-stage conceptual pipeline, from raw signal to human alert



Continuous feedback loop: outcomes from each alert refine model accuracy for future detections.

Core AI & ML Techniques

The analytical layer that turns raw sensor data into a fire signal



CNN Image Classification

Convolutional neural networks scan satellite and drone imagery to recognize smoke plumes and flame patterns.



Multi-Sensor Data Fusion

Temperature, humidity, gas and visual data are combined for a single, more reliable fire indicator.



Anomaly Detection

Models flag deviations from normal environmental baselines that suggest an emerging fire event.



Predictive Risk Modeling

Historical fire, terrain and weather data estimate fire-spread probability across a region.

Data Sources & Sensing Layer

Where the system observes the forest from



Satellite Imagery

Wide-area thermal and optical imaging captures large-scale hotspots and smoke plumes across remote regions.



Ground IoT Sensors

Distributed nodes measure temperature, humidity, smoke particles and gas levels at canopy level.



Drone Surveillance

Aerial units provide close-range, on-demand imagery for high-risk or hard-to-reach zones.



Weather Data Feeds

Wind speed, humidity and temperature trends contextualize fire risk and likely spread direction.

SENSING ALTITUDE STACK

Space — Satellites

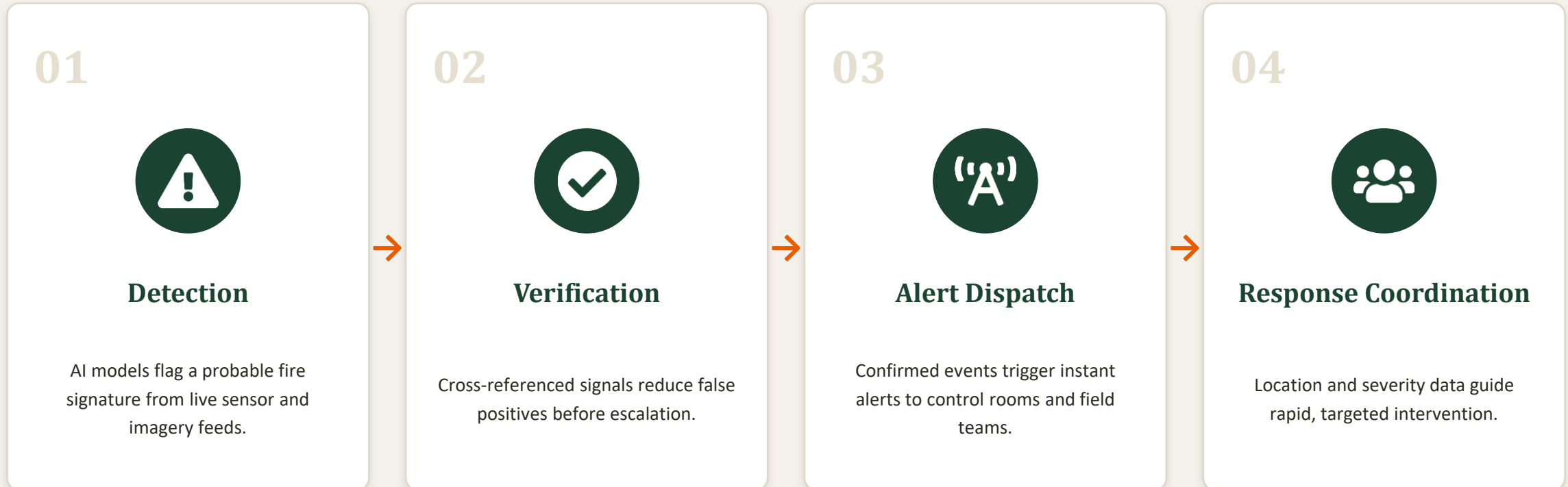
Air — Drones

Ground — IoT Sensors

Layered coverage ensures no blind spots across scale or terrain.

Early Warning & Alert Workflow

From confirmed detection to coordinated response





Thank You

Protecting forests through earlier, smarter detection

Questions & Discussion